

AIQT Milestone 14 Build Specification v0.2

Agent Operating Discipline and Planning Hardening

Field	Value
Product	AIQT CLI
Specification	Milestone 14 Build Specification
Version	v0.2
Milestone	M14 - Agent Operating Discipline and Planning Hardening
Status	Build-ready review candidate
Depends on	M1-M13 complete, including M13 Planning-Phase Design System Skill
Primary evidence	Moodboard Landing Page Builder dogfood and Feedback Board dogfood findings F053-F057

1. Source Alignment

This document is derived from the AIQT MVP Specification v0.6, Technical Architecture Specification v0.3, and the completed milestone sequence M1 through M13. M14 is a hardening milestone informed by the Moodboard Landing Page Builder and Micro SaaS Feedback Board dogfoods.

The governing workflow remains: state -> graph -> packet -> checkpoint -> review/export -> issue/recovery controls -> planning/design guidance. M14 does not add a new workflow stage. It tightens how agents operate existing stages.

AIQT remains a local state engine. It does not call Claude, OpenAI, Playwright MCP, Hue, transitions.dev, shadcn installers, or any external API. It does not modify source code automatically.

2. v0.2 Change Summary

- Moves packet audit metadata into the authoritative historical layer: agent_packet.created runlog event data.
- Keeps state.lastAgentPacket as an optional latest-packet mirror only, not the sole audit record.
- Requires agent-packet export to read latest valid packet audit metadata from runlog when available.
- Clarifies recovery guidance scope: M14 adds driver guidance only; graph validate/repair behavior remains owned by M12.
- Clarifies that lastAgentPacket examples extend the existing AgentPacketMetadata shape and must preserve format/sourceCommand/contentHash fields.
- Retains M14 v0.1 scope: no new commands, no new runlog event types, no schema version bump expected, no full packet body persistence.

3. Dogfood Findings Addressed

- F053 - Component-system preference such as shadcn/ui was not enforced. M14 response: Strengthen design-system planner and packet guidance to enforce declared component-system preferences.
- F054 - Agent can create a nested application folder or drift from implementation root. M14 response: Add working-directory discipline guidance to driver and packets.
- F055 - External tools such as Playwright MCP use Claude workspace cwd rather than existingRepositoryPath. M14 response: Explicitly state AIQT control root, implementation root, command cwd, and artifact expectations.
- F056 - Agent packet export cannot audit whether design/working-directory guidance was present. M14 response: Persist renderedSections/guidanceFlags in agent_packet.created runlog data and export them.
- F057 - Agent used reset/reimport rather than M12 repair controls after graph mistake. M14 response: Add driver recovery discipline that prefers M11/M12 controls before reset/reimport.

4. Objective

M14 hardens agent operating discipline and planning guidance after M13. It makes AIQT better at instructing external agents where to work, how to respect component-system preferences, how to avoid nested-root drift, how to use repair controls, and how to audit packet guidance after the fact.

5. In Scope

- Enhance aiqt prompt driver with working-directory and recovery-discipline guidance.
- Enhance aiqt next packet rendering with working-directory discipline when existingRepositoryPath is set or differs from AIQT control root.
- Enhance design-system planning and packet guidance so declared component-system preferences such as shadcn/ui are enforceable.
- Extend agent_packet.created runlog event data with renderedSections and guidanceFlags.
- Optionally mirror latest packet audit metadata into state.lastAgentPacket while preserving existing metadata fields.
- Update agent-packet export so it can report historical packet audit flags from runlog metadata.
- Add tests for no command-surface expansion, no new runlog event types, no mutation from read-only commands, and packet audit history.

6. Out of Scope

- No new public commands.
- No package publishing or npm release readiness work.
- No automatic shadcn/ui installation.
- No calls to Claude Design, Hue, transitions.dev, Playwright MCP, or external APIs.
- No full packet body persistence.
- No source-code modification by AIQT.
- No graph repair apply mode, broad replanning, or new recovery commands.
- No schema version bump unless implementation proves unavoidable.

7. Behavioral Position in the Workflow

M14 modifies prompt and packet guidance within the existing workflow. It does not change capture, planning ingestion, work graph validation, checkpoint schema, review semantics, issue lifecycle, or graph mutation behavior.

- aiqt prompt driver: Adds operating-root discipline and repair-before-reset guidance.
- aiqt prompt plan: Strengthens component-system preference enforcement through existing M13 design-system planner.
- aiqt prompt checkpoint: May remind the agent to report cwd/artifacts and component-system deviations in existing checkpoint fields.
- aiqt next: Adds working-directory discipline and component-system guidance when relevant; records audit metadata in runlog.
- aiqt export agent-packet: Shows latest historical packet audit metadata from runlog when available.
- aiqt graph validate / repair: No behavior change in M14; M12 owns these commands.

8. Working-Directory Discipline

When existingRepositoryPath is set, AIQT guidance must distinguish the AIQT control root from the implementation repository root. This is especially important when Claude Code or Playwright MCP is launched from the wrong folder.

8.1 Guidance block

```
## Working Directory Discipline
```

```
AIQT control root:  
<cwd containing .aiqt>
```

```
Implementation root:
```

<project.existingRepositoryPath>

Rules:

- Run AIQT commands from the AIQT control root.
- Run application commands from the implementation root.
- Run browser / Playwright MCP checks from the implementation root.
- Make application source changes only inside the implementation root unless the packet explicitly says otherwise.
- Do not create nested application folders under the AIQT control root.
- Do not create .playwright-mcp or browser artifacts under the AIQT control root when an implementation root is configured.
- Report changed files relative to the implementation root in checkpoint output.

8.2 Rendering rules

- Include the block in aiqt prompt driver when existingRepositoryPath is present.
- Include the block in aiqt next packets when existingRepositoryPath is present or differs from the AIQT control root.
- If no existingRepositoryPath is configured, do not invent one. Instead warn that the implementation root is assumed to be the AIQT control root.

9. Component-System Preference Enforcement

M13 introduced planning-phase design-system guidance. M14 hardens it so a declared component-system preference such as shadcn/ui cannot be silently replaced by hand-rolled primitives.

9.1 Detection and inputs

- Reuse the existing M13 UI-heavy detector and design-system planner helpers. Do not create a second detector.
- Read component-system preferences from project.context.technologyPreferences, project.context.architectureNotes, requirements, decisions, and existing repository evidence such as components.json.
- Supported explicit preference examples: shadcn/ui, shadcn-ui, Radix primitives, custom Tailwind primitives. M14 only needs strong behavior for shadcn/ui because it is the current dogfood finding.

9.2 shadcn/ui enforcement guidance

If project context declares shadcn/ui:

- Planning guidance must require a shadcn/ui setup or integration work unit.
- Design-system foundation must distinguish tokens from component-library implementation.
- Agent packets for UI work must say to use shadcn/ui primitives or wrappers, not duplicate hand-rolled Button/Input/Card primitives.
- Custom primitives are allowed only when explicitly justified in checkpoint notes.
- Skills plan should keep shadcn-ui integration detection/recommendation behavior from M10/M13.

9.3 Acceptance criteria template

For UI work units in a shadcn/ui project, include or recommend criteria such as:

- components.json or equivalent shadcn/ui configuration exists.
- Core controls use shadcn/ui-based primitives or project wrappers around them.
- No duplicate hand-rolled Button/Input/Card primitives exist unless justified.
- Tailwind tokens and shadcn/ui CSS variables are aligned.
- Keyboard focus states and disabled/loading states are preserved.

10. Recovery Discipline Guidance

M14 does not change aiqt graph validate or aiqt graph repair behavior. M12 owns graph validation and repair suggestions. M14 only improves driver guidance so external agents prefer existing M11/M12 controls before destructive reset/reimport.

When workflow state looks wrong:

- If an issue needs lifecycle handling, use aiqt issue list / update / promote.
- If a dependency type is wrong, use aiqt dependency update <id> --type <type> --reason "...".
- If graph health is unclear, use aiqt graph validate and aiqt graph repair --dry-run.
- If a checkpoint result was classified incorrectly, use aiqt checkpoint amend.
- Reset/reimport is a last resort for early disposable plans only; do not use it once meaningful checkpoints exist.

11. Packet Audit Metadata

The authoritative historical packet audit record is the `agent_packet.created` runlog event. `state.lastAgentPacket` may mirror the latest packet metadata but must not be the only storage location for audit metadata.

11.1 Runlog event data extension

```
{
  "eventType": "agent_packet.created",
  "actor": "cli",
  "data": {
    "packetId": "PKT-012",
    "workUnitId": "WU012",
    "milestoneId": "M005",
    "contentHash": "sha256:...",
    "renderedSections": [
      "scope",
      "context",
      "constraints",
      "acceptance",
      "validation",
      "designGuidance",
      "workingDirectoryDiscipline",
      "componentSystemGuidance"
    ],
    "guidanceFlags": {
      "includesDesignGuidance": true,
      "includesWorkingDirectoryDiscipline": true,
      "includesComponentSystemGuidance": true,
      "includesRecoveryGuidance": false
    }
  }
}
```

This is an additive extension to the existing `agent_packet.created` event data. It does not introduce a new event type.

11.2 lastAgentPacket mirror

If the implementation adds `renderedSections/guidanceFlags` to `state.lastAgentPacket`, it must preserve the full existing `AgentPacketMetadata` shape, including existing fields such as `id`, `workUnitId`, `milestoneId`, `createdAt`, `contentHash`, `format`, and `sourceCommand`. The illustrative snippets in this spec extend the existing shape; they do not replace it.

11.3 Export behavior

aiqt export agent-packet should read the latest valid `agent_packet.created` runlog event matching the exported packet ID when available. If audit metadata exists, the export must show the rendered section list and guidance flags. If metadata is missing for old packets, the export must degrade gracefully and state that guidance audit metadata is unavailable for that packet.

Example export addition:

```
## Packet Audit Metadata
```

Rendered sections:

- Design Guidance: yes
- Working Directory Discipline: yes
- Component System Guidance: yes
- Recovery Guidance: no

Audit source: `agent_packet.created` runlog event

12. Command Contract

- aiqt prompt driver: Adds working-directory and recovery discipline guidance. Mutation: no.
- aiqt prompt plan: Strengthens component-system enforcement through existing M13 design planner. Mutation: no.
- aiqt prompt checkpoint: May add wording to report cwd/artifacts and component-system deviations. Mutation: no.
- aiqt next: Renders additional guidance sections and records audit metadata in `agent_packet.created` runlog data. Mutation: existing packet mutation only.
- aiqt export agent-packet: Displays packet audit metadata from runlog when available. Mutation: exports only.
- aiqt graph validate / repair: No M14 behavior change. Mutation: no change.

13. Exit Codes

M14 adds no new commands and should not add new exit-code behavior. Existing command outcomes remain authoritative. Invalid state or malformed canonical files continue to return exit code 3 through existing command pathways. Read-only prompt/skills/export behavior remains read-only except for export file writes where already established.

14. Data and Schema Contracts

- No AIQT_SCHEMA_VERSION bump expected because all additions are optional and additive.
- StateModel may add optional fields to lastAgentPacket metadata only if existing schema patterns allow it.
- Runlog schema must allow optional renderedSections and guidanceFlags on agent_packet.created data.
- Old runlog events without audit metadata must remain valid.
- Old state files without guidanceFlags/renderedSections must remain valid.

15. Repository Implementation Scope

Likely new or modified files:

```
src/workflow/agent-operating-discipline.ts
src/workflow/component-system-preferences.ts
src/workflow/design/design-guidance-rules.ts
src/workflow/design/ui-heavy-detection.ts
src/services/agent-packet-service.ts
src/services/agent-packet-template.ts
src/services/export-service.ts
src/services/export-templates.ts
src/templates/prompts/driver.prompt.template.ts
src/templates/prompts/plan.prompt.template.ts
src/templates/prompts/checkpoint.prompt.template.ts
src/schema/runlog-event.schema.ts
src/schema/state.schema.ts (only if latest-packet mirror is added)
src/state/runlog-store.ts
```

```
tests/unit/agent-operating-discipline.test.ts
tests/unit/component-system-preferences.test.ts
tests/integration/agent-packet-audit-metadata.test.ts
tests/integration/export-agent-packet-audit.test.ts
tests/integration/prompt-driver-operating-discipline.test.ts
tests/integration/prompt-plan-component-system.test.ts
```

Implementation may choose different filenames if existing repository conventions are clearer. The critical requirement is shared helper reuse, not exact names.

16. Testing Scope

16.1 Unit tests

- Working-directory guidance renders correct control root and implementation root.
- Component-system preference detection recognizes shadcn/ui from technology preferences and repository evidence.
- shadcn/ui guidance does not appear for projects that explicitly prefer custom primitives only.
- Packet audit flags are computed from rendered sections, not hard-coded.
- Old packet metadata without audit fields remains valid.

16.2 Integration tests

- aiqt prompt driver includes working-directory discipline when existingRepositoryPath is configured.
- aiqt prompt driver includes repair-before-reset guidance.
- aiqt prompt plan requires shadcn setup/integration guidance when shadcn/ui is declared.
- aiqt next writes renderedSections/guidanceFlags into agent_packet.created runlog data.
- aiqt export agent-packet displays packet audit metadata from runlog.
- Export gracefully handles older packet events with no audit metadata.
- No new commands are registered.

- No new runlog event types are introduced.

17. Runtime and Package Impact

- No new runtime dependency expected.
- No network access.
- No external tool execution.
- No package manager invocation.
- No change to package publishing behavior.

18. Validation Commands

Preferred validation commands, if pnpm scripts are available:

```
pnpm typecheck
pnpm lint
pnpm test
pnpm build
pnpm validate
```

Known fallback for the current repository environment when pnpm validate is blocked by ignored build scripts:

```
node_modules/.bin/tsc --noEmit
node_modules/.bin/eslint .
node_modules/.bin/vitest run
node_modules/.bin/tsc
```

19. Definition of Done

- All M1-M13 tests continue to pass.
- M14 tests prove packet audit metadata is stored historically in agent_packet.created runlog event data.
- agent-packet export can audit guidance flags for the latest exported packet from runlog.
- Working-directory discipline appears in driver and packet surfaces when existingRepositoryPath is set.
- Component-system preference enforcement appears in plan and packet guidance for declared shadcn/ui projects.
- Recovery guidance appears in driver prompt and does not imply M14 changes to M12 graph commands.
- No new public commands, no new event types, no schema version bump, and no external execution.

20. Required Implementation Report

The implementing agent must return:

- Branch name and commit hash.
- Files changed summary, including created/modified source and test files.
- Whether schema changed and whether AIQT_SCHEMA_VERSION changed.
- Whether commands changed and whether any new command was registered.
- Whether runlog event types changed; if not, confirm only agent_packet.created data was extended.
- Validation results and final test count.
- How packet audit metadata is stored and exported.
- Deviations from spec, if any.
- Deferred/unresolved items, if any.
- Working tree status after commit.

21. Agent Execution Packet

Starting point:

- main contains M1-M13 complete and merged.

- Create branch: feat/m14-agent-operating-discipline-hardening.

Task:

Implement M14 v0.2 exactly as specified.

Key constraints:

- No new commands.
- No external tool/API calls.
- No full packet body persistence.
- Add packet audit metadata to agent_packet.created runlog data.
- Keep state.lastAgentPacket audit fields as an optional mirror only.
- Preserve existing AgentPacketMetadata fields.
- Reuse M13 UI-heavy detection and design guidance helpers.
- Do not modify M12 graph validate/repair behavior.

Validation:

Run the repository validation commands and return the required implementation report.